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Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

Software Project Report

on the topic **Football Players' Statistics and Transfer Market: Math Perspective**

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Аннотация

Проект «Статистика футболистов и трансферный рынок: математический анализ» направлен на изменение оценки стоимости футболистов за счет использования методов машинного обучения. Анализируя обширные массивы данных с сайта Transfermarkt, проект нацелен на разработку надежной и точной модели прогноза рыночной стоимости игроков. Работа затрагивает такие области, как: предварительная обработка данных, регрессионный анализ и обнаружение аномалий с целью повышения эффективности трансферного рынка и снижения финансовых рисков.

Проект включает в себя несколько важных этапов: сбор и анализ данных с использованием таких инструментов, как BeautifulSoup, предварительная обработка для обеспечения качества данных и кодирование функций для соответствующих атрибутов игрока. RandomForestRegressor используется для прогнозирования значений трансферной стоимости игрока, алгоритм оптимизирован с использованием фреймворка настройки гиперпараметров Optuna. Кроме того, проект выявляет аномалии ценообразования и устраняет присущий им дисбаланс классов с помощью таких методов, как SMOTE и RandomForestClassifier.

Первоначальные результаты показали расхождения между прогнозируемой и фактической рыночной стоимостью, но эти расхождения находились в приемлемых пределах. Несмотря на улучшения за счет методов оптимизации и повторной выборки, надежность модели в реальных приложениях остается несколько ограниченной из-за дисбаланса наборов данных. В частности, существенного улучшения требует прогнозирование ценовых аномалий. Будущая работа будет сосредоточена на совершенствовании этих моделей и изучении более сложных методов, позволяющих лучше справляться с несоответствиями данных и повышать точность как прогнозирования рыночной стоимости, так и обнаружения аномалий.

В итоге этот проект направлен на создание основанной на данных системы оценки стоимости игроков, предлагающей ценные сведения для клубов, агентов, инвесторов и любителей спорта, тем самым способствуя принятию более обоснованных решений в футбольной индустрии.

Abstract

This project, "Football Players' Statistics and Transfer Market: Math Perspective," aims to revolutionise the valuation of football players by leveraging machine learning techniques. By analyzing comprehensive datasets from the Transfermarkt website, the project seeks to develop robust predictive models that can accurately forecast player market values. The initiative addresses key areas including data preprocessing, regression analysis, and anomaly detection, aiming to enhance transfer market efficiency and mitigate financial risks.

The project encompasses several critical stages: data collection and parsing using tools like BeautifulSoup, preprocessing to ensure data quality, and feature encoding for relevant player attributes. A RandomForestRegressor is employed to predict player transfer values, optimised using the hyperparameter tuning framework Optuna. As well as that, the project identifies pricing anomalies and addresses the inherent class imbalance with techniques like SMOTE and RandomForestClassifier.

Initial results indicated gaps between predicted and actual market values, although these discrepancies were within acceptable limits. Despite improvements through optimisation and resampling techniques, the model's reliability in real-world applications remains somewhat limited due to dataset imbalances. Particularly, the prediction of pricing anomalies requires significant enhancements. Future work will focus on refining these models and exploring more sophisticated methods to better handle data disparities and improve the accuracy of both market value predictions and anomaly detection.

This project ultimately aims to provide a data-driven framework for player valuation, offering valuable insights for clubs, agents, investors, and sports enthusiasts, thereby contributing to more informed decision-making in the football industry.

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Basic Terms and Definitions

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalise to unseen data, and thus perform tasks without explicit instructions. Recently, artificial neural networks have been able to surpass many previous approaches in performance.

Sports analytics are collections of relevant historical statistics that can provide a competitive advantage to a team or individual. Through the collection and analysis of these data, sports analytics inform players, coaches and other staff in order to facilitate decision making both during and prior to sporting events.

Data preprocessing can refer to manipulation, filtration or augmentation of data before it is analysed, and is often an important step in the data mining process. Data collection methods are often loosely controlled, resulting in out-of-range values, impossible data combinations, and missing values, amongst other issues.

Transfermarkt is a German-based website owned by Axel Springer SE that has footballing information, such as scores, results, statistics, transfer news, and fixtures. The website has scores, results, transfer news, fixtures, and player values. Despite the player values, along with some other facts, being estimates, researchers from the Centre for Economic Performance have found that the "rumours" of player transfers are largely accurate.

In statistics, **mean absolute error (MAE)** is a measure of errors between paired observations expressing the same phenomenon. Examples of Y versus X include comparisons of predicted versus observed, subsequent time versus initial time, and one technique of measurement versus an alternative technique of measurement. MAE is calculated as the sum of absolute errors (i.e., the Manhattan distance) divided by the sample size:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

XGBoost is an open-source software library which provides a regularising gradient boosting framework for Python and other programming languages. From the project description, it aims to provide a «Scalable, Portable and Distributed Gradient Boosting (GBM, GBRT, GBDT) Library».

Random Forests or Random Decision Forests is an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time. For classification tasks, the output of the random forest is the class selected by most trees. For regression tasks, the mean or average prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set.

1. Introduction

The field of sports analytics and performance prediction has seen a significant transformation owing to advancements in machine learning. With the use of machine learning methods, sports teams can easily get useful information and then take decisions that are well based on data. This infusion of data science into sports has redefined player assessment and game strategies, as well as team leadership — which can set apart one professional football team from another.

Aimed at reaching a broad spectrum of beneficiaries in the football universe, the project "Football Players' Statistics and Transfer Market: Math Perspective" targets clubs, agents, investors and lovers of sports. The essence of the project involves coming up with a predictive model that can look at player performance from historical data and market trends— thus arriving at reasonable and value-based player valuations. In this light, this particular venture holds significance to enriching transfer market efficiency: also risk assessment and delineation of player development strategy.

The player transfer market plays a role, in the football landscape shaping the future of football clubs. Accurate player valuations are key for making investment decisions and navigating the market effectively. Traditional methods often rely on judgments. With the integration of advanced machine learning models there is a shift towards more precise and unbiased estimations of players future performance and market value.

To enhance market insights the initiative uses datasets from the Transfermarkt website. A source provides extensive information on football players. Through regression analysis the data is processed to understand how various performance metrics influence player valuations. Also, anomaly detection techniques are applied to detect trends or discrepancies in valuations providing insights, into the dynamics of the transfer market.

To sum up, "Football Players' Statistics and Transfer Market: Math Perspective" is a noteworthy development in the field of sports analytics. In the long run, this initiative hopes to transform player values by utilising large datasets and machine learning to provide a solid foundation for data-driven decision-making in the football business.

1.1. Project Purpose

Building a predictive model for the transfer value of football players.

1.2. Project Objectives

To study materials related to the project theme provided by the project supervisor to enhance understanding. To implement data collection and parsing from the Transfermarkt website using the BeautifulSoup library for efficient and targeted data extraction. This included parsing performance data from national team games.

To be responsible for identifying and encoding key player performance features, including player positions, trophies and national team data, essential for market value assessment using data processing techniques.

Developing regression models for predicting player transfer values, utilising the RandomForestRegressor from the scikit-learn library. To improve predictive accuracy, optimise model parameters using Optuna.

To identify pricing anomalies and share these insights with a colleague to assist in training his model, providing valuable data for deeper analysis and model improvement.

2. Review and comparative analysis of sources and analogues

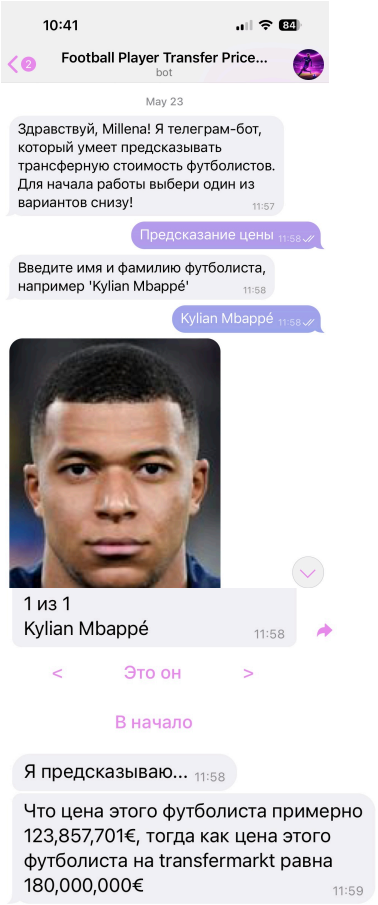
Our project, "Football Players' Statistics and Transfer Market," focuses on predictive modeling of football players' transfer values, while "TransferLab: Analytics FC" targets player recruitment, valuation, and performance benchmarking. Both leverage data for enhanced decision-making in football but serve distinct purposes and audiences.

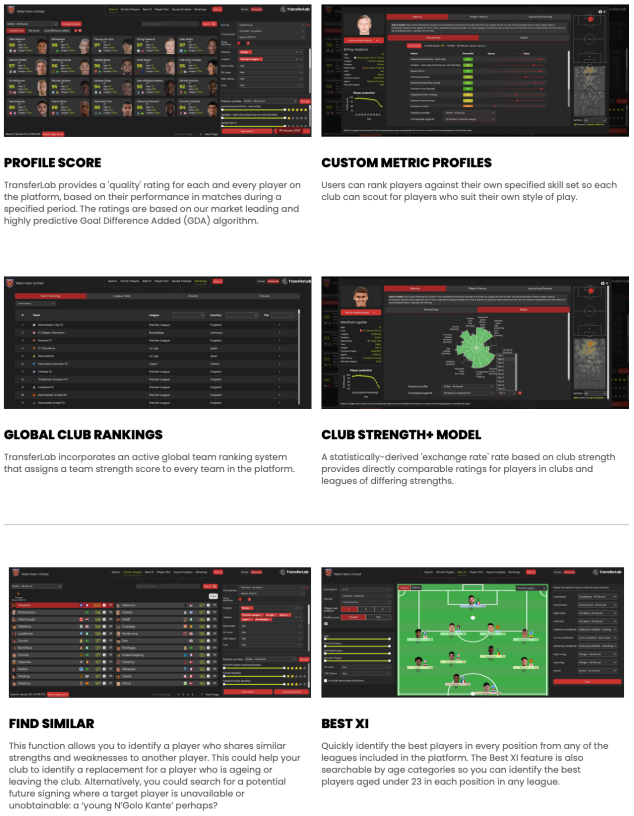
Our project uses Transfermarkt for extensive player statistics, transfers, and market values, employing regression methods to predict transfer values. TransferLab utilizes various data sources and advanced statistical models for professional recruitment and performance analysis.

We plan to develop a user-friendly interface for clubs, agents, investors, and enthusiasts, aiming for wider accessibility. TransferLab already offers a professional-grade interface primarily for football clubs and analysts.

Our primary focus is on predicting transfer values, aiding clubs and agents in the transfer market, while TransferLab offers broader functionalities related to player performance and recruitment. Although we do not provide comprehensive performance analysis beyond transfer value prediction, TransferLab offers detailed analysis. Both models consider external factors such as player age, injury history, and market demand.

In summary, both projects enhance decision-making in football through data but differ in application and audience. Our project emphasizes broader accessibility and transfer value prediction, while TransferLab provides a comprehensive tool for professional recruitment and performance analysis.





3. Description of the Functional and Non-functional Requirements

3.1 Functional requirements

Data Collection and Parsing, Preprocessing, Feature Encoding and Statistical Analysis:

1. The system must automatically collect player data from the Transfermarkt website.
2. It should use the bs4 library to parse and extract relevant data efficiently.
3. The system must preprocess raw data to make it suitable for analysis (including handling missing values, outliers, and normalising data).
4. It should encode categorical features and possibly engineer new features for better model performance.
5. The system must perform statistical analysis to identify patterns and correlations in the data.

Development of Predictive Models:

1. The system must develop regression models (linear and more advanced models) using scikit-learn to predict player transfer values.
2. It should implement classification models to predict the anomaly in market values.
3. The system must consider external factors like age, injury history, and market demand in the analysis.

3.2 Non-functional requirements

- The system must have a high level of data integrity and accuracy. All data collected and processed should be verified for correctness and relevance.
- The system must ensure data security, especially in handling sensitive player information.
- The system should be designed to support the English language in order to ensure universal accessibility and understanding.
- The system should consistently provide accurate and reliable predictions and analysis.
- The system should be compatible with various platforms, ensuring accessibility across different browsers.
- The system must be scalable to accommodate increasing amounts of data and more complex models in the future.
- The system should be designed for easy maintenance and updates, allowing for future enhancements and bug fixes.
- Comprehensive documentation and customer support should be provided to assist users.

4. Project Implementation

This is an individual project functioning within the scope of a larger collaborative effort involving six students. Together, we parse data to compile a complete dataset as the foundational asset for the entire project. Upon successful aggregation and preprocessing of the data, the project transitions from a cooperative to an individualistic stage. Each student will then undertake an independent journey to apply machine learning methodologies to the dataset. The individual work will involve developing distinct ML models suited to our respective research questions within the broader theme of the project.

4.1 Parsing National Team Career Data

Parsing data from the Transfermarkt in the context of my project aims to gather detailed national team career data of football players. This process involves making HTTP requests, parsing HTML content, and extracting specific data points to build a comprehensive dataset.

```
urls = [
    'https://www.transfermarkt.com/neymar/profil/spieler/68290', # Неймар играет за сборную Бразилии
    'https://www.transfermarkt.com/cristiano-ronaldo/profil/spieler/8198', # CR7 за Португалию
    'https://www.transfermarkt.com/rafa-marin/profil/spieler/662261', # Молодой Испанец за U21
    'https://www.transfermarkt.com/brahim-traore/profil/spieler/743517', # Б.Трауре молодой француз из U20
    'https://www.transfermarkt.com/axel-witsel/profil/spieler/35047', # Витсель завершил игры за сборную Белгии
    'https://www.transfermarkt.com/karim-benzema/profil/spieler/18922' # Бензема, которого выгнали из сборной Франции

def NationalTeamCareer(url):
```

The data extraction process begins with an HTTP GET request to the Transfermarkt player profile URL using the `requests` library to fetch HTML content. The content is parsed using `BeautifulSoup` to locate sections containing national team career data. Key data points, such as national team name and player status, are extracted and encoded into numerical values: `0` for Not Playing, `1` for playing in a youth national team, and `2` for playing in the senior national team.

We utilize `requests` for HTTP requests, `BeautifulSoup` for HTML parsing, and regular expressions for text processing. Player statuses are identified by analyzing HTML classes, and team names are cleaned to remove age group indicators.

Data regarding national team career for each individual player is taken out. Moreover, other performance statistics are collected from the specific detailed page of their national team profile by using the `scrape_national_team_stats` function. This includes extracting metrics such as the count of national matches, goals scored, assists made, yellow cards received, and red cards shown.

A structured format like a DataFrame is prepared with all details of every player when the data is extracted and compiled. This dataset is significant to be comprehensive in building strong predictive models that would take into account player performance metrics— among other determinants — to establish the worth of players in markets, as well as anomaly detection.

	National Team	National Status	National Matches	National Goals	National Assists	National Yellow Cards	National Red Cards
0	Brazil	2	128	79	59	32	1
1	Portugal	2	205	128	46	30	-
2	Spain	1	6	-	-	1	-
3	France	1	3	-	-	1	-
4	Belgium	0	130	12	8	14	2
5	France	0	97	37	20	1	-

4.2 Preprocessing Data for Analysis

In our project preparing the data is crucial to make sure the dataset is organised and set for, in depth analysis and modelling. When it comes to preparing the data I focus on a tasks; updating national team information managing trophies and organising player positions.

Data Cleaning: This process entails fixing mistakes resolving discrepancies and filling in missing information to guarantee precision and dependability for deriving insights and making well informed choices.

Data Transformation: Refining raw data, for analysis involves converting structures consolidating information and generating variables.

National Team Preprocessing:

This part preprocessing involves changing the status codes to show if a player is active in their National Team currently. The following statuses are updated;

-1: The player has never been called to the national team.

0 : The player is not currently active in the national team.

1 : The player is part of a youth national team.

2 : The player is an active member of the senior national team.

This is an introduction of a rank for each National Team for additional context provision. Teams are ranked according to their performances with higher ranks being given to the best-performing sides. In other words, teams like Brazil, Argentina and France have been ranked the most. Afterward, the dataset is sorted with respect to the column called National Team Ranking.

Trophies Preprocessing

The trophies preprocessing takes into account the prizes at both individual and club levels. They each have specific weights according to their importance, where highly competitive events like UEFA Champions League and domestic league titles bear more weight. The process involves the following:

- Parsing Individual Trophies: The string of trophies won by every player is broken down into individual trophies. For each player, scores are determined using preset weights, which gives the total, average, maximum and minimum scores in a single season.

- Parsing Club Trophies: Similarly, every club achievement is identified and given its appropriate weight. These may encompass historical as well as recent accomplishments so as to reflect the player's contribution towards their team's success. The club score computation follows the same approach as that of individual awards thus ensuring an all-rounded assessment of career accolades per player.

Positions Preprocessing

The preprocessing of player positions involves expanding the position column into multiple columns using one-hot encoding. Each unique position is converted into a binary column, indicating whether a player has played in that position. To simplify the analysis, positions are grouped into broader categories:

- *Defender Group*: Includes positions like Defender, Centre-Back, and Sweeper.
- *Midfielder Group:* Encompasses Attacking Midfield, Central Midfield, Defensive Midfield, and general midfield roles.
- *Striker Group*: Covers positions such as Attack, Centre-Forward, and Second Striker.
- *Winger Group*: Includes Left Midfield, Left Winger, Right Winger, Right Midfield, Left-Back, and Right-Back.

These grouped columns provide a clearer view of a player's positional versatility and primary roles on the field, facilitating more effective analysis and modelling.

The preprocessing steps outlined above are essential for structuring the data into a format suitable for further analysis and modelling. By systematically handling national team statuses, trophy achievements, and player positions, we ensure a comprehensive dataset ready for robust predictive models.

4.3 Price Prediction Model

The machine learning component of the project initially focused on developing a DecisionTreeRegressor model incorporating a broad spectrum of football players' performance metrics. These metrics ranged from goals and assists to the number of trophies, as well as other pertinent on- and off-field factors impacting a player's market value. This initial effort aimed to uncover complex factors crucial in setting player valuations, focusing on various metrics including injuries and international performances. However, the initial results, obtained using k-fold cross-validation, were unsatisfactory, with a significant Mean Absolute Error (MAE) of 11,055,686.08, highlighting a considerable gap between predicted and actual market values. This discrepancy was primarily attributed to the constrained dataset and the model's rudimentary structure.

```
MAE scores:
10079472.84345048
10123162.939297125
11177076.677316293
11854246.794871794
12044471.153846154

Average MAE: 11055686.08
Standard Deviation of MAE: 830990.07
```

Acknowledging these shortcomings, we implemented a comprehensive strategy to augment our dataset, enhance the model's sophistication, and apply a more deliberate method in weighting each feature.

Model Development and Optimization

The relevant features selected for the model include a wide range of player attributes such as age, height, appearances, goals, assists, substitutions, cards, minutes played, injury statistics, club

statistics, trophies, and positions. These features are essential for capturing the multifaceted aspects of player performance and market value.

To develop the predictive model, we used the **RandomForestRegressor** from the scikit-learn library. The initial model was trained and evaluated, and its performance was measured using MAE.

Recognising the need for further optimisation, the project utilised Optuna, a hyperparameter optimisation framework, to fine-tune the model parameters such as the number of estimators, maximum features, and maximum depth. The objective function for Optuna was defined to minimise the MAE.

```
▾ RandomForestRegressor
RandomForestRegressor(max_depth=20, max_features=0.9406014627190983,
                      n_estimators=418, random_state=42)
```

```
best_predictions = best_rf.predict(X_test)
best_mae = mean_absolute_error(y_test, best_predictions)
print(f'Best MAE: {best_mae}')
```

```
Best MAE: 1104183.7807345195
```

Finally, the model was trained on the entire dataset to make predictions for all players and saved for future use. The optimized RandomForestRegressor, with parameters such as max_depth=20, max_features=0.94, and n_estimators=418, was fitted to the training data and evaluated on the test data, resulting in an improved MAE.

This process ensured that our price prediction model is robust and capable of providing valuable insights into player valuations, which are crucial for making informed decisions in the football transfer market. The comprehensive approach to feature selection, model training, and hyperparameter optimisation laid a solid foundation for accurate and reliable player market value predictions.

4.4 Anomaly Value Prediction

The anomaly value prediction in this project aims to identify discrepancies by comparing the predicted player transfer price with the actual Transfermarkt price. An anomaly is defined when the price difference exceeds a threshold of approximately 3.6 million euros. This model is intended to help avoid market imbalances, where a mediocre player is priced like a top player. Unfortunately,

	full_name	age	height	citizenship	current_club	current_price	predicted_price	price_difference
0	Joško Gvardiol	22.0	1.85	Croatia	Manchester City	80000000	7.004785e+07	9.952153e+06
1	Nathan Benjamin Aké	29.0	1.80	Netherlands Cote d'Ivoire	Manchester City	40000000	4.555263e+07	5.552632e+06
3	Kyle Andrew Walker	33.0	1.78	England	Manchester City	13000000	1.697847e+07	3.978469e+06
4	Matheus Luiz Nunes	25.0	1.83	Portugal Brazil	Manchester City	55000000	4.960048e+07	5.399522e+06
5	Bernardo Mota Veiga de Carvalho e Silva	29.0	1.73	Portugal	Manchester City	80000000	7.570335e+07	4.296651e+06

the dataset is highly imbalanced, with 315 anomalies and about 15,000 non-anomalies, making accurate modelling challenging.

Initially, I observed that most anomaly players were from top clubs and top 5 leagues, suggesting that high prices might be influenced by marketing campaigns. Using an XGBClassifier model, the initial results showed an accuracy of 0.97986 but an F1 score of 0.403669, primarily due to the class imbalance.

To address this, I applied the Synthetic Minority Over-sampling Technique (SMOTE) and used a RandomForestClassifier to enhance the model's performance. Despite these efforts, the model still faced limitations and is not recommended for real-life application due to its inconsistencies.

The optimisation process for the XGBClassifier model involved tuning hyperparameters with Optuna. This process suggested various parameters such as booster type, regularization terms (lambda and alpha), max depth, eta (learning rate), subsample, and colsample_bytree. The best trial achieved an F1 score of 0.45045045045045046, indicating some improvement but still not sufficient.

```
XGBClassifier
XGBClassifier(alpha=0.019450565236428755, base_score=None, booster='dart',
              callbacks=None, colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8067053527182989, device=None,
              early_stopping_rounds=None, enable_categorical=False,
              eta=0.18416621851547876, eval_metric=None, feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, lambda=0.017883665686658928,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=4,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=None, ...)
```

```
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))
```

```
Accuracy: 0.9798636926889716
F1 Score: 0.40366972477064217
```

Applying SMOTE to the training data and retraining with RandomForestClassifier, the model achieved an accuracy of 0.97986 and improved the F1 score to 0.5185185185185185. While these improvements are notable, the model's reliability in real-world scenarios remains questionable due to the inherent imbalance in the dataset.

```
Accuracy: 0.9798636926889716
F1 Score: 0.5185185185185185
```

While this project has made strides in predicting anomalies in player transfer values, the significant class imbalance poses a challenge.

The models developed show promise but require further refinement and perhaps additional data to enhance their real-world applicability and reliability. Future work could explore alternative balancing techniques or more sophisticated models to better handle the data imbalance and improve the accuracy and reliability of anomaly predictions in the football transfer market.

Conclusion

The project "Football Players' Statistics and Transfer Market: Math Perspective" successfully developed a predictive model for football player transfer values using machine learning techniques. Comprehensive datasets from the Transfermarkt website were used to build models that forecast player market values. Key stages included data collection, preprocessing, feature encoding, and model development, focusing on regression analysis and anomaly detection.

A RandomForestRegressor was employed to predict transfer values, optimised with Optuna, and anomaly detection was performed using an XGBClassifier. Despite addressing class imbalance with SMOTE and RandomForestClassifier, the model's reliability in real-world applications remains limited due to dataset imbalances.

Further Recommendations

1. Incorporate additional data sources for a more comprehensive dataset.
2. Advanced feature engineering - Explore additional influential features like player popularity and off-field activities.
3. Investigate advanced machine learning models and continuously fine-tune hyperparameters.
4. Develop a user-friendly interface for wider audience accessibility, providing clear visualisations and actionable insights.

References

(1) Git Repository of the project on the implementation of a model for predicting the transfer value of football club players.

(2) Wikipedia Machine Learning provides a broad overview of machine learning, its applications, and methodologies, offering a foundational understanding for implementing ML techniques in this project.

(3) Transfermarkt provides comprehensive datasets on player statistics, transfers, and market values, essential for data collection.

(4) BeautifulSoup and Requests libraries are used for data extraction. BeautifulSoup's documentation guides HTML parsing, while Requests handles HTTP requests to fetch data from Transfermarkt.

(5) Scikit-learn offers a variety of algorithms and techniques for implementing machine learning models, crucial for this project's regression and classification tasks.

(6) Medium articles provide practical exercises and real-world applications of machine learning in predicting market values, demonstrating the potential of these models to enhance transfer market efficiency.

(7) The ML HSE Course by E. Sokolov provide theoretical foundations and practical insights into the algorithms used in this project.

(8) Comparative analysis of TransferLab: Analytics FC focuses on player recruitment, valuation, and performance benchmarking for professional use. In contrast, my project emphasises predicting transfer values and making the results accessible to a broader audience.